Information Systems Planning and Acquisition

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Information Systems Planning

- Information systems planning is a complex process which needs collection and analysis of vast amounts of information about organisational business, markets, policy and environment, etc.
- Formal and comprehensive analysis could improve the quality of decision-making and the effectiveness of ISP (Das et al).
- Another definition: Information systems planning (ISP) is a process of defining objectives for organisational computing and identifying suitable potential information technology applicable to the company.

Information Systems Acquisition

- The acquisition of information systems can either involve external sourcing or rely on internal development or modification.
- With today's highly developed IT industry, companies tend to acquire information systems and services from specialised vendors.

Information Systems Analysis

- IS analysis is the process of assessing the needs of an organisation and determining the type of information systems required to meet those needs.
- It involves:
  - **Understanding Business Needs:** Identifying key business functions and how IS can support these processes.
  - **Gap Analysis:** Comparing current systems with desired outcomes to identify deficiencies.

Information Systems Analysis

- **Requirements Gathering:** Collecting detailed functional and non-functional requirements from stakeholders.
- **Feasibility Study:** Analysing the technical, operational, and financial feasibility of proposed solutions.
- Key considerations include:
  - **Technical Feasibility:** Can the technology be implemented with the available infrastructure?
  - **Operational Feasibility:** Will the system meet business needs and fit within the organisation's culture?
  - **Economic Feasibility:** Is the solution cost-effective?

Information Systems Planning

- IS planning is a strategic activity that aligns the development of IS with the overall business strategy.
- Key aspects include:
  - **Strategic Alignment:** Ensuring that IS initiatives are in line with business objectives.
  - **Resource Allocation:** Budgeting and allocating resources (people, technology, finances) for IS development.
  - **Prioritisation:** Ranking IS projects based on their strategic importance, potential impact, and resource availability.

Information Systems (IS) Planning

- **Risk Management:** Identifying and mitigating risks associated with IS implementation.
- IS planning typically follows a structured methodology:
  - **Strategic IS Planning:** Long-term planning (3-5 years) that looks at future trends and technology advancements.
  - **Tactical IS Planning:** Short-term planning (1-2 years) focused on specific projects and immediate needs.

Software Acquisition Options

- Organisations have several options when acquiring software.
- It can be developed internally which is referred to as proprietary software, or it can be purchased off-the-shelf, also referred to as packaged software.

Software Acquisition Options

- **Custom Software Development:** Building software in-house or outsourcing its development to fit specific organisational needs.
  - **Pros**: Tailored to business processes.
  - **Cons:** Expensive, time-consuming, and high risk.
- **Off-the-Shelf Software**: Purchasing pre-built software that is ready to use.
  - **Pros**: Faster implementation, cost-effective.
  - **Cons**: Limited customisation.

Software Acquisition Options

- **Open Source Software:** Free software where source code is available for modification.
  - **Pros:** Cost-effective, flexible.
  - **Cons:** Requires technical expertise for customisation
- **Software as a Service (SaaS):** Renting software via the cloud on a subscription basis.
  - **Pros:** Lower upfront costs, scalability, accessibility.
  - **Cons:** Less control over data and security concerns.

Outsourcing

- Outsourcing involves contracting external vendors to handle certain IT functions or entire IS projects.
- Outsourcing models include:
  - **Full Outsourcing:** A vendor manages the entire IT infrastructure and services.
  - **Selective Outsourcing:** Specific IT functions (e.g., data storage, helpdesk support) are outsourced.

Outsourcing

- **Offshore Outsourcing:** Outsourcing to companies located in another country.
  - **Advantages**:
    - Cost savings.
    - Access to specialised expertise.
    - Focus on core business operations.
  - **Disadvantages**:
    - Loss of control.
    - Potential quality issues.
    - Security and confidentiality concerns.

Project Management

- Project management in IS ensures that information systems projects are delivered on time, within budget, and meet the defined objectives.
- Key components include:
  - **Project Initiation**: Defining project goals, scope, and stakeholders.
  - **Planning**: Developing a project plan that outlines tasks, timelines, resources, and budget.

Project Management

- Key components include:
  - **Execution:** Carrying out the project according to the plan, while managing teams and resources.
  - **Monitoring and Controlling:** Tracking project progress and making necessary adjustments to keep it on course.
  - **Closing:** Finalising the project, ensuring objectives are met, and delivering the final product.
- Popular project management methodologies in IS:
  - **Waterfall:** A linear, sequential approach to software development.
  - **Agile:** An iterative, flexible approach that emphasises collaboration and adaptability.

IS People Aspects

- The success of any IS project depends on the people involved.
- Key roles and responsibilities in an IS team include:
  - **Chief Information Officer (CIO):** Oversees the IS strategy and ensures alignment with business objectives.
  - **Project Manager:** Manages IS projects, ensuring they are completed on time and within budget.
  - **Systems Analyst:** Acts as a bridge between business needs and technical requirements, helping design solutions.

IS People Aspects

- Key roles and responsibilities in an IS team include:
  - **Developers:** Build and implement software solutions based on technical specifications.
  - **End Users:** The individuals who interact with the system and provide feedback for continuous improvement.

IS People Aspects

- Human Factors to Consider:
  - **Training and Support:** Ensuring users are adequately trained to use the new systems.
  - Change Management: Managing resistance and ensuring smooth transitions to new systems.
  - User Involvement: Involving users in the development process to ensure the system meets their needs.